

Advanced (Habanero)

Q1. Solve the inequality $|x| < 3$.

- (A) $(-\infty, -3) \cup (3, \infty)$
 (B) $(-3, 3)$
 (C) $(-\infty, 3)$
 (D) $[3, \infty)$
 (E) $(-\infty, -3] \cup [3, \infty)$

Q2. Graph the solution set of $|x - 2| \geq 1$ on a number line.

- (A) $(-\infty, 1] \cup [3, \infty)$
 (B) $[1, 3]$
 (C) $(-\infty, 1) \cup (3, \infty)$
 (D) $[2, \infty)$
 (E) $(-\infty, 2] \cup [2, \infty)$

Q3. The absolute value of a number x is less than 5. What is the solution set in interval notation?

- (A) $(-5, 5)$
 (B) $[-5, 5]$
 (C) $(-\infty, -5) \cup (5, \infty)$
 (D) $(-5, \infty)$
 (E) $(-\infty, 5)$

Q4. What is the solution set of $|2x - 3| > 2$?

- (A) $(-\infty, \frac{1}{2}) \cup (\frac{5}{2}, \infty)$
 (B) $(-\infty, \frac{1}{2}] \cup [\frac{5}{2}, \infty)$
 (C) $(-\infty, \frac{1}{2}) \cup [\frac{5}{2}, \infty)$
 (D) $(-\infty, \frac{1}{2}] \cup (\frac{5}{2}, \infty)$
 (E) $(-\infty, \frac{1}{2}) \cup (\frac{5}{2}, \infty)$

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 (E) $(-\infty, \frac{1}{2}) \cup (\frac{5}{2}, \infty)$

Q5. Solve $|x + 1| \leq 2$.

- (A) $[-3, 1]$
 (B) $(-\infty, -3) \cup (1, \infty)$
 (C) $[-1, 3]$
 (D) $(-3, 1)$
 (E) $(-\infty, -1] \cup [3, \infty)$

Q6. The temperature in a city is within 3°C of 20°C . Express this as an absolute value inequality.

- (A) $|T - 20| < 3$
 (B) $|T - 20| > 3$
 (C) $|T - 20| = 3$
 (D) $T - 20 < 3$
 (E) $T - 20 > 3$

Q7. A company packages coffee in bags that weigh within 0.1 pounds of 1 pound. Express this as an absolute value inequality.

- (A) $|x - 1| \leq 0.1$
 (B) $|x - 1| \geq 0.1$
 (C) $|x - 1| = 0.1$
 (D) $x - 1 \leq 0.1$
 (E) $x - 1 \geq 0.1$

Q8. Solve $|x| > 2$.

- (A) $(-\infty, -2) \cup (2, \infty)$
 (B) $(-2, 2)$
 (C) $(-\infty, 2)$
 (D) $[2, \infty)$
 (E) $(-\infty, -2] \cup [2, \infty)$

Open Ended Questions (FRQ)

Q9. Model the situation 'the distance from a point x to -4 is less than 7' as an absolute value inequality and solve for x . Provide your solution in interval notation.**Q10.** A bakery sells cookies in boxes that weigh within 0.05 pounds of 0.5 pounds. Express this situation as an absolute value inequality and graph the solution set on a number line. Provide your solution with interval notation.

Chef's Tips

- Check calculations carefully.
- Use interval notation for solutions.
- Graph solutions on a number line.